

WEEK 8: STUDENT PERFORMANCE DASHBOARD REPORT

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1. Objective

The objective of this week was to build an interactive Student Performance Analytics Dashboard using the cleaned dataset from Week 7. The dashboard answers these questions:

- What is the overall academic performance?
- Which subject has the highest and lowest average?
- Does test preparation relate to performance?
- How do parental education and lunch type relate to scores?
- Which groups have higher or lower pass rates?

2. Dataset and Data Model

The Week 7 Power BI file, with its cleaned and transformed data and DAX measures, was reused. The data model and measures were checked before building the visuals. The measures used are Average Math Score, Average Reading Score, Average Writing Score, Overall Average Score and Overall Pass %.

3. KPI Cards

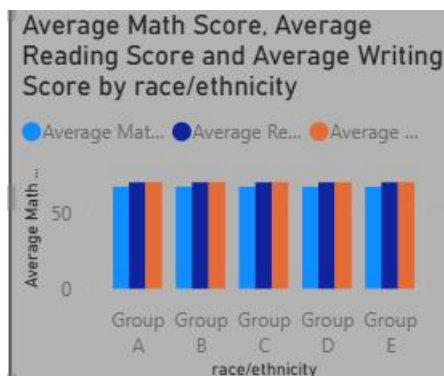
KPI	Value
Average Math Score	67.36
Average Reading Score	70.03
Average Writing Score	70.15
Overall Average Score	207.54 (total of the three subjects)
Overall Pass %	97%

These values are for the selected filter (Group D). Math is the lowest-scoring subject, and Writing is the highest by a small margin over Reading.



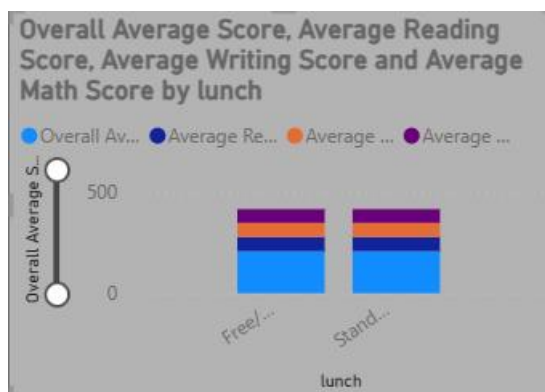
4. Subject Performance by Race/Ethnicity

This chart compares the three subject averages across Groups A to E. The bars are close in height for all groups, so performance differences between groups appear small.



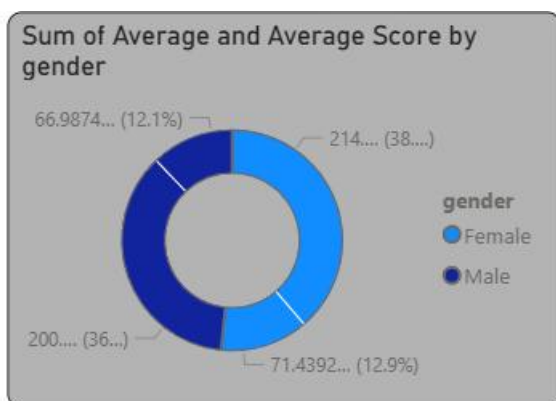
5. Performance by Lunch Type

This visual compares score measures for Free/Reduced and Standard lunch groups. Lunch type is used as an indicator of socioeconomic background.



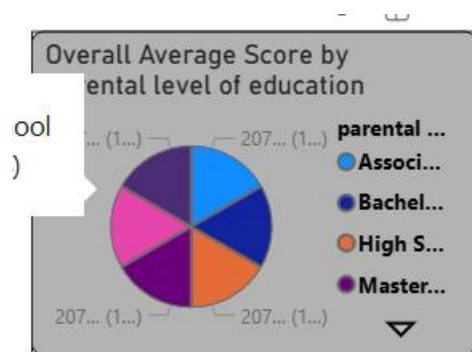
6. Performance by Gender

The donut chart shows the female segments (214, 71.4) slightly larger than the male segments (200, 67.0) in the selected group.



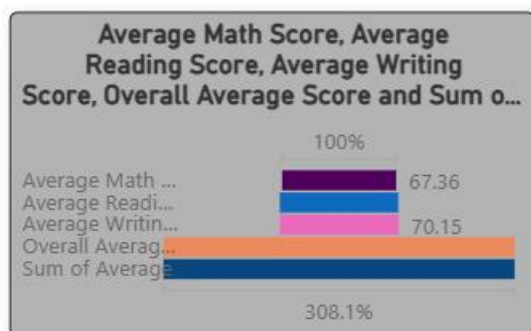
7. Parental Education

The slices are almost equal in size, so no strong difference by parental education level is visible in the selected data.



8. Subject Comparison Summary

This chart summarises the subject averages (Math 67.36, Writing 70.15) next to the overall measures.



9. Slicers and Interactivity

1. The dashboard has slicers for race/ethnicity, test preparation course and lunch type. Selecting a value updates all visuals.
- 2 This chart compares average scores and pass % for students who completed test preparation and those who did not. Write your values here: Completed = ____, None = ____.
3. This chart shows the count or percentage of students in each grade.

race/ethnicity

☐ Group A

☐ Group B

☐ Group C

☒ Group D

☐ Group E

test preparation cour...

☐ Completed

☐ None

lunch

☐ Free/Reduced

☐ Standard

13. Key Insights

Insight 1: Math is the weakest subject.

Observation: Math has the lowest average score.

Evidence: Math is 67.36, Reading is 70.03 and Writing is 70.15.

Interpretation: In the selected group, students score about 2.7 to 2.8 points lower in math than in the language subjects.

Insight 2: The pass rate is very high.

Observation: Almost all students pass.

Evidence: Overall Pass % is 97%.

Interpretation: Only about 3% of students fall below the pass mark, so average scores and grades are better for comparing groups.

Insight 3: Subject averages are similar across groups.

Observation: Groups A to E have similar bar heights in all three subjects.

Evidence: The clustered column chart shows no group with a dramatically different profile.

Interpretation: Differences between race/ethnicity groups are small in this dataset.

This is an association, not a cause.

Insight 4: Female students show slightly higher values.

Observation: The female segments are larger than the male segments.

Evidence: Female 214 and 71.4 against Male 200 and 67.0.

Interpretation: Female students show higher values on the measures shown, within the selected group.

Insight 5: Parental education shows no clear difference.

Observation: The pie chart slices are nearly equal.

Evidence: All slices show similar values (about 207).

Interpretation: Parental education does not show a visible effect on average score in the selected data.

Insight 6: Test preparation.

Observation: Students who completed test preparation have ____ average score compared with ____ for those who did not.

Interpretation: Test preparation completion is associated with different observed performance levels in this dataset. No causal claim is made.

15. Conclusion

The Student Performance Dashboard summarises academic performance through KPIs, subject comparisons, demographic analysis, test preparation analysis and drill-through. Math is the lowest-scoring subject, the pass rate is about 97%, and the differences between groups are small. All findings are observed associations and do not prove causation.